Checkpoint1:

```
[INFO  0,2] worker: thread 2: count 73000, sleeping
[sched 0,2] sched: switch to scheduler pid(2)
[sched 0,-1] scheduler: switch to proc pid(3)
[INFO  0,3] worker: thread 3: count 74000, sleeping
[sched 0,3] sched: switch to scheduler pid(3)
[sched 0,-1] scheduler: switch to proc pid(4)
[INFO  0,4] worker: thread 4: count 75000, sleeping
[sched 0,4] sched: switch to scheduler pid(4)
[sched 0,-1] scheduler: switch to proc pid(5)
[INFO  0,5] worker: thread 5: count 76000, sleeping
[sched 0,5] sched: switch to scheduler pid(5)
[sched 0,-1] scheduler: switch to proc pid(6)
[INFO  0,6] worker: thread 6: count 77000, sleeping
[sched 0,6] sched: switch to scheduler pid(6)
[sched 0,-1] scheduler: switch to proc pid(7)
[INFO  0,7] worker: thread 7: count 78000, sleeping
[sched 0,7] sched: switch to scheduler pid(7)
[sched 0,-1] scheduler: switch to proc pid(8)
[INFO  0,8] worker: thread 8: count 79000, sleeping
[sched 0,8] sched: switch to scheduler pid(8)
[sched 0,-1] scheduler: switch to proc pid(9)
[INFO  0,9] worker: thread 9: count 80000, sleeping
[sched 0,9] sched: switch to scheduler pid(9)
[sched 0,-1] scheduler: switch to proc pid(2)
[WARN  0,2] worker: thread 2: exiting
[sched 0,2] sched: switch to scheduler pid(2)
[sched 0,-1] scheduler: switch to proc pid(3)
[ERROR 0,3] kernel_trap: kerneltrap: not from supervisor mode
[ERROR 0,3] kernel_trap: =========== Kernel Panic ===========
sstatus : 0x8000000200006020
- SUM:0, SPP:U, SPIE:1, SIE: 0
scause  : 0x0000000000000002
- Interrupt:0, Code:2
sepc    : 0x000000008020197c
stval   : 0x00000000100024f3
sip     : 0x0000000000000000
- Pending: Software:0, Timer:0, External:0
sie     : 0x0000000000000220
- Enabled: Software:0, Timer:1, External:1
satp    : 0x0000000000000000
kernel trapframe at 0x000000008023ce60
ra: 0x0000000080202364   sp: 0x000000008023ce60   gp: 0x0000000000000000  tp: 0x0000000000000000
t0: 0x0000000000000000   t1: 0x000000008023cf48   t2: 0x0000000000001000  s0: 0x000000008023cf80
s1: 0x0000000000000000   a0: 0x00000000000001f4   a1: 0x000000000000000a  a2: 0x0000000000000005
a3: 0x00000000004c4b40   a4: 0x000000000afa2cec   a5: 0x0000000080202364  a6: 0x00000000802062c0
a7: 0x0000000000000000   s2: 0x0000000000000000   s3: 0x00000000000003e8  s4: 0x0000000000000003
s5: 0x0000000080205ad8   s6: 0x00000000802054b0   s7: 0x0000000080206300  s8: 0x0000000000000000
s9: 0x0000000080238d98  s10: 0x0000000000000000  s11: 0x0000000000000000  t3: 0x0000000000000010
t4: 0x0000000080046dda   t5: 0x000000000000000f   t6: 0x0000000000000027

[PANIC 0,3] os/trap.c:89: kernel panic
```

可以看到有报错信息：kerneltrap: not from supervisor mode 和 kernel panic

同时可以看到 8 个工作线程能轮流运行计数

Checkpoint2：

Spp bit 会在执行 sret 指令时进行切换，当从陷阱处理程序返回时，硬件会根据 SPP 的值恢复特权级（0 表示用户模式，1 表示监督模式），然后**将** SPP **设置为 0**。这意味着每次 SRET 执行后，SPP 都会被清零。

```
[sched 0,9] sched: switch to scheduler pid(9)
[sched 0,-1] scheduler: switch to proc pid(2)
[WARN  0,2] worker: thread 2: exiting
[sched 0,2] sched: switch to scheduler pid(2)
[sched 0,-1] scheduler: switch to proc pid(3)
[WARN  0,3] worker: thread 3: exiting
[sched 0,3] sched: switch to scheduler pid(3)
[sched 0,-1] scheduler: switch to proc pid(4)
[WARN  0,4] worker: thread 4: exiting
[sched 0,4] sched: switch to scheduler pid(4)
[sched 0,-1] scheduler: switch to proc pid(5)
[WARN  0,5] worker: thread 5: exiting
[sched 0,5] sched: switch to scheduler pid(5)
[sched 0,-1] scheduler: switch to proc pid(6)
[WARN  0,6] worker: thread 6: exiting
[sched 0,6] sched: switch to scheduler pid(6)
[sched 0,-1] scheduler: switch to proc pid(7)
[WARN  0,7] worker: thread 7: exiting
[sched 0,7] sched: switch to scheduler pid(7)
[sched 0,-1] scheduler: switch to proc pid(8)
[WARN  0,8] worker: thread 8: exiting
[sched 0,8] sched: switch to scheduler pid(8)
[sched 0,-1] scheduler: switch to proc pid(9)
[WARN  0,9] worker: thread 9: exiting
[sched 0,9] sched: switch to scheduler pid(9)
[sched 0,-1] scheduler: switch to proc pid(1)
[INFO  0,1] init: thread 2 exited with code 22
[INFO  0,1] init: thread 3 exited with code 23
[INFO  0,1] init: thread 4 exited with code 24
[INFO  0,1] init: thread 5 exited with code 25
[INFO  0,1] init: thread 6 exited with code 26
[INFO  0,1] init: thread 7 exited with code 27
[INFO  0,1] init: thread 8 exited with code 28
[INFO  0,1] init: thread 9 exited with code 29
[INFO  0,1] init: all threads exited, count 80

[INFO  0,1] init: init ends!
[PANIC 0,1] os/proc.c:225: init process exited
```

Ckeckpoint3：

```
[WARN  2,5] worker: thread 5: exiting
[WARN  1,12] worker: thread 12: exiting
[WARN  3,16] worker: thread 16: exiting
[WARN  3,3] worker: thread 3: exiting
[INFO  2,1] init: thread 3 exited with code 23, expected 23
[WARN  3,7] worker: thread 7: exiting
[INFO  1,6] worker: thread 6: count 149000, sleeping
[WARN  3,13] worker: thread 13: exiting
[WARN  0,4] worker: thread 4: exiting
[INFO  3,1] init: thread 4 exited with code 24, expected 24
[WARN  0,8] worker: thread 8: exiting
[INFO  3,1] init: thread 5 exited with code 25, expected 25
[WARN  0,15] worker: thread 15: exiting
[WARN  1,11] worker: thread 11: exiting
[WARN  3,9] worker: thread 9: exiting
[WARN  2,10] worker: thread 10: exiting
[WARN  0,14] worker: thread 14: exiting
[WARN  3,6] worker: thread 6: exiting
[INFO  3,1] init: thread 6 exited with code 26, expected 26
[INFO  3,1] init: thread 7 exited with code 27, expected 27
[INFO  3,1] init: thread 8 exited with code 28, expected 28
[INFO  3,1] init: thread 9 exited with code 29, expected 29
[INFO  3,1] init: thread 10 exited with code 30, expected 30
[INFO  3,1] init: thread 11 exited with code 31, expected 31
[INFO  3,1] init: thread 12 exited with code 32, expected 32
[INFO  3,1] init: thread 13 exited with code 33, expected 33
[INFO  3,1] init: thread 14 exited with code 34, expected 34
[INFO  3,1] init: thread 15 exited with code 35, expected 35
[INFO  3,1] init: thread 16 exited with code 36, expected 36
[INFO  3,1] init: all threads exited, count 149866

[INFO  3,1] init: init ends!
[PANIC 3,1] os/proc.c:225: init process exited
[PANIC 0,-1] os/trap.c:47: other CPU has panicked
[PANIC 2,-1] os/trap.c:47: other CPU has panicked
[PANIC 1,-1] os/trap.c:47: other CPU has panicked
```

这是第 20 次运行的结果

作业完成时间：

从开始到结束一共三个小时，期间包含休息时间，满打满算应该两个小时能完成

指引建议：

1. 对于第二个检查点，其实真正的问题在于，在连续调用的环节导致当前 trap 对应的 sepc/sstatus 被后续别的 trap 路径覆盖了；但是我没从

hint4 的提示中读出来这个意味（也有可能是我的理解问题），如果不能直接说明而导致作业太简单，可以尝试提醒同学们关注调用前后某些值的变化